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Ma'aden Engineering Manual

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1. INTRODUCTION

1.1 Scope

This document provides the minimum requirements for geotechnical investigation during the various stages of development for all Ma'aden projects.

1.2 Purpose

The purpose of this specification is to set the minimum engineering standard for geotechnical investigation for all Ma'aden projects.

2. KEY TERMS AND DEFINITIONS

Contractor	An entity, engaged or approved by the Owner, entrusted with a contract to perform work or cause work to be performed.
Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates.
Geotechnical	Pertaining to the nature, condition and physical properties of the earth's crust (whether soil or rock and including water and gases therein) which affect its performance in civil engineering and building works.
Geotechnical Data	Information collected and assembled during the geotechnical investigation. It can be 'factual', i.e. materials, statistics and properties that can be seen, measured or identified by means of accepted or standardized criteria, classifications and tests, or 'interpretative', i.e. information derived from factual data using accepted and proven techniques, or from reasonable judgment exercised in the assessment of geological conditions or processes evident at the Site.
Geotechnical Investigations	The process of evaluating the geotechnical character of a Site in the context of existing or proposed works or land usage.
In-Situ	In its original/natural place.
Monitoring	The activity of sensing, observing, measuring, and recording changes over a period of time.
Site	Geographical location where Work is to be performed.
Soil Classification	A systematic method of categorizing soils according to their possible engineering behavior.
Work	The entire work and Services and all things which the Contractor is or may be required to execute, supply and do under the Contract having regard to the Scope of Project, or to cause to be executed,

supplied or done, for the purposes of constructing and Commissioning the Plant.

CHT	Cross Hole Seismic Test.
CPT	Cone Penetration Test
DCP	Dynamic Cone Penetrometer
DMT	Flat Plate Dilatometer Test
ERT	Electrical Resistivity Test
PMT	Pressuremeter Test
PT	Percolation Test
RQD	Rock Quality Designation
SPT	Standard Penetration Test
SRT	Seismic Refraction Test
VST	Vane Shear Test

3. REFRENCED DOCUMENTS

This specification requires compliance with the provisions of the latest adopted revision of the relevant standards and codes of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable.

3.1 Statutory Regulations

All Work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia (including but not limited to directives of the High Commission for Industrial Security, Royal Commission regulations applicable to the Site, regulations of the Presidency of Meteorology and Environment, Saudi Building Code and other applicable codes and regulations). The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

The Work shall conform to the following Standards and Codes of Practice.

3.2.1 Local Codes and Standards

SBC 303 Soil and Foundations Requirements

3.2.2 International Codes and Standards

3.2.2.1 American Society for Testing and Materials (ASTM)

ASTM D420	Standard Guide to Site Characterization for Engineering, Design and Construction Purposes
ASTM D698	Standard Test methods for Laboratory Compaction Characteristics of Soil Using Standard Effort
ASTM D1452	Standard Practice for Soil Exploration and Sampling by Auger Borings
ASTM D1556	Standard Test Method for Density and Unit Weight of Soil in Place by the Sand-Cone Method (AASHTO T191)
ASTM D1557	Standard Test methods for Laboratory Compaction Characteristics of Soil Using Modified Effort
ASTM D1586	Standard Test Method for Standard Penetration Test (SPT) and Split-Barrel Sampling of Soils
ASTM D1587	Standard Practice for Thin-Walled Tube Sampling of Soils for Geotechnical Purposes
ASTM D1883	Standard Test methods for CBR (California Bearing Ratio) of Laboratory-Compacted Soils
ASTM D2113	Standard Practice for Rock Core Drilling and Sampling of Rock for Site Investigation
ASTM D2166	Standard test Method for Unconfined Compressive Strength of Cohesive Soil
ASTM D2435	Standard Test Method for One–Dimensional Consolidation Properties of Soils
ASTM D2487	Standard Test Method for Classification of Soils for Engineering Purposes
ASTM D2488	Standard Practice for Description and Identification of Soils (Visual-Manual Procedure)

ASTM D2573	Standard Test Method for Field Vane Shear Test in Cohesive Soil
ASTM D2850	Standard Test Method for Unconsolidated-Undrained Triaxial Compression Test on Cohesive soils
ASTM D3080	Standard Test Method for Direct Shear Test of Soils Under Consolidated Drained Conditions
ASTM D3441	Standard Test Method for Mechanical Cone Penetration Tests of Soil
ASTM D4220	Standard Practices for Preserving and Transporting Soil Samples
ASTM D4253	Standard Test Method for Maximum Index and Unit Weight of Soils using a Vibratory Table
ASTM D4254	Standard Test Method for Maximum Index and Unit Weight of Soils and Calculation of Relative Density
ASTM D4318	Standard Test Method for Liquid Limit, Plastic Limit, and Plasticity Index of Soils
ASTM D4428	Standard Test Method for Crosshole Seismic Testing
ASTM D4719	Standard Test Method for Pre-Bored Pressuremeter Testing in Soils
ASTM D5092	Standard Practice for Design and Installation of Ground Water Monitoring Wells
ASTM D5334	Standard Test Method for Determination of Thermal Conductivity of Soil and Soft Rock by Thermal Needle Probe Procedure
ASTM D5434	Standard Guide for Field Logging of Subsurface Exploration of Soil and Rock
ASTM D5753	Standard Guide for Planning and Conducting Borehole Geophysical Logging
ASTM D5777	Standard Guide for Using the Seismic Refraction Method for Subsurface Investigation
ASTM D5778	Standard Test Method for Electronic Friction Cone and Piezocone Penetration Testing of Soils
ASTM D6032	Standard Test Method for Determining Rock Quality Designation (RQD) of Rock Core

ASTM D6034	Standard Test Method (Analytical Procedure) for Determining the Efficiency of a Production Well in a Confined Aquifer from a Constant Rate Pumping Test
ASTM D6151	Standard Practice for Using Hollow-Stem Augers for Geotechnical Exploration and Soil Sampling
ASTM D6635	Standard Test Method for Performing the Flat Plate Dilatometer
ASTM D6951	Standard Test Method for Use of the Dynamic Cone Penetrometer in Shallow Pavement Applications
ASTM D7015	Standard Practices for Obtaining Intact Block (Cubical and Cylindrical) Samples of Soils
ASTM D7380	Standard Test Method for Soil Compaction Determination at Shallow Depths Using 5-lb (2.3 kg) Dynamic Cone Penetrometer
ASTM D7400	Standard Test Methods for Downhole Seismic Testing
ASTM G57	Standard Test Method for Field Measurement of Soil Resistivity Using the Wenner Four-Electrode Method
AASHTO T26	Standard Specification for Quality of Water to be used in Concrete
AASHTO T290	Standard Method of Test for Determining Water Soluble Sulfate Ion Content in Soil
AASHTO T291	Standard Method of Test for Determining Water Soluble Chloride Ion Content in Soil

3.2.3 Alternative Codes

Alternative codes may be used for Geotechnical Investigation if approved by the Employer.

3.3 Ma'aden Standards

MD-101-SMEM-EG-CV-SPC-0001	Earthworks
MD-101-SMEM-EG-CV-SPC-0003	Roadworks
MD-101-SMEM-EG-CV-SPC-0008	Topographical Survey
MD-101-SMPM-PM-QA-PEI-0001	Contractor Quality Requirements

4. GENERAL REQUIREMENTS

4.1 Personnel for Geotechnical Investigations

The Contractor shall assign qualified personnel to conduct the geotechnical investigations. Geotechnical investigations should be carried out by professional specialists such as engineering geologists, geological engineers, geotechnical engineers, geologists or civil engineers with education and experience in the type of work envisaged. Technicians shall be experienced in the proper operation of all equipment they are assigned to operate. The Employer reserves the right to reject any person from working on the project for any reason Employer deems reasonable.

4.2 Equipments/Instruments

The Contractor shall utilize the appropriate equipments and instruments based on the requirements of geotechnical investigations. All equipments/instruments shall be in good working conditions and calibrated before the commencement of site work.

4.3 Safety

The Contractor shall take all necessary actions to ensure the safety of personnel, equipment and properties throughout the field investigation and shall include mobilization/demobilization of all necessary equipments, technical personnel, skilled and unskilled labors on Site.

4.4 Environment

The Contractor shall take all necessary actions to manage impact to the environment during geotechnical investigation activities on Site. The Contractor shall seek information on environmental requirements as and when necessary.

4.5 Project Execution Plan

The Contractor shall submit a Project Execution Plan (PEP) as specified in the Scope of Work document. The PEP shall provide a means by which all project objectives will be met. The PEP shall include detail hold points for the project including but not limited to the following:

- Method Statement
- Desktop Review
- Draft Report.

4.6 Method Statement

The Contractor shall submit a detailed method statement for the execution of the Work for approval prior to the commencement of the Work.

4.7 Scope of Work

The following items shall be considered during the Contractor's planning and shall form part of the scope of Work.

- A description of the project

- The purpose and objectives of the geotechnical investigation
- Type of structures to be constructed
- Approximate loads
- Final grade and/or anticipated cut or fill
- Depth of basements, utility trenches, pits, ponds, etc
- Information on Site access, ID and work permit requirements
- Proponent contact and phone number
- Field Test Schedule including the number, type, and depths of field tests to be conducted
- The objectives of the laboratory test program (specific number and type of laboratory tests need not be specified)
- A description of any special tests or requirements
- A location plot plan including the locations of the existing and proposed facilities and the locations of the field tests
- A completed checklist of required Geotechnical Services.

5. DESKTOP REVIEW

5.1 Assessment of Project Requirements

The Contractor shall review the basic (physical, chemical and engineering) parameters of the project. The review shall include the project location, orientation of the project, locations of the structures, delineation of project, cut and fill areas and any restrictions on construction activities and type of products (specially waste material) based upon natural conditions, project requirements and environmental conditions at the project Sites.

5.2 Work Permits

Local government agencies require permits for the field exploration that disturbs the vegetation and topographic features. The requirements vary depending on local engineering and environmental ordinances. The Contractor shall be responsible to obtain all the required permits, before the start of the field investigations. The Contractor shall identify the responsible government agencies and adequate allowance for timely approval of permits shall be planned.

The Contractor shall investigate and verify adequacy regarding previous land use of the Site which could affect the project in any way. This is important particularly in areas where there have been underground works e.g. worked out ballast pits, quarries, old brick fields, mines, mineral workings, etc. The possibility of damage to the structures, sewers, conduits and drainage system by subsidence shall also be investigated.

The Contractor shall obtain dig permits before the commencement of the site investigation.

5.3 Review of Available Data

The Contractor shall gather all useful data on previous geotechnical investigations in the area or adjacent projects or local government/city/council libraries and literature. The Contractor shall, as a minimum, investigate the following sources:

- Prior subsurface investigations (historical data) at or near the project Site
- Prior construction and records of structural performance problems at the Site (i.e. pile length, driveability problems, rock slides, excessive seepage, unpredicted settlement, and other information)
- Geological survey maps, reports and publications
- State flood zone maps
- Karst (limestone sinkholes) environments
- Earthquake data, seismic hazards maps, fault maps
- State, city, and county road maps
- Aerial photographs
- Remote sensing Images
- Site plans showing locations of ditches, driveways, culverts, utilities, and pipelines
- Maps of streams, rivers and other water bodies to be crossed by bridges, culverts, etc. including bathymetric data.

6. FIELD EXPLORATION AND MAPPING

6.1 General

Following the collection and evaluation of available information from the sources identified in Section 5.3, the Contractor shall plan the field exploration program.

The Contractor shall carry out the walk over survey for the field exploration. The walk-over survey shall be conducted by a qualified geotechnical engineer or geologist to obtain first-hand knowledge of the field conditions, surface soils and topology, and correlate this information with previous data. The Contractor shall note the following information during the walk over survey:

- Visual inspection of adjacent structures shall be completed to ensure the protection of these buildings is accounted for in the design. Observations including photographic references of this inspection shall be part of the geotechnical report whenever possible.

- On water crossings, banks should be inspected for scour and the streambed inspected for evidence of soil deposits not previously indicated
- Note any feature that may affect the boring program, such as accessibility, structures, overhead utilities, signs of buried utilities, or property restrictions
- Note any feature that may assist in the engineering analysis, such as the angle of any existing slopes and the stability of any open excavations or trenches
- Any drainage features, including signs of seasonal water levels
- Any features that may need additional borings or probing such as muck pockets.

Where rock is outcropped on the surface, information on rock type and the structural features of rock mass shall be measured. The Contractor shall assign qualified geologists or geotechnical engineers with the relevant rock mechanics experience to identify the rock types. The basic rock mapping information such as measurement of the inclination (dip) and orientation (dip direction) and structural features such as bedding planes, cleavage and joint planes shall be recorded.

The Contractor shall conduct a risk assessment for underground cavities when Greenfields projects (new facilities or infrastructure constructed on previously undeveloped land) are located where underlying limestone formations are present.

6.2 Material/Consumables

The Contractor shall supply all material, equipment and consumables required for the excavation of test pits and/or drilling of boreholes, including sample bags, drilling mud, cement, sample cylinders.

The Contractor shall also provide sample storage during sampling.

7. DRILLING AND SAMPLING OF SOIL AND ROCK

7.1 General

Drilling shall be performed at approved locations to obtain information about the sub-soil profile, its nature, strength and to collect soil samples for strata identification and conducting laboratory tests. The drilling shall be carried out in accordance with the provisions of relevant standards referenced in Section 3 and as per this specification.

All drilling holes shall extend up to the depths shown on the location plans or as directed by the Employer.

The collection of samples shall be carried out at regular intervals and at change of strata or as directed by the Employer. Sufficient disturbed and undisturbed samples shall be preserved for conducting tests in the laboratory. Water levels in the borehole shall be carefully recorded after stabilization of the water level in the borehole and reported.

On completion of the borehole, the portion drilled in soil shall be backfilled with sand or

sand-cement mortar, unless otherwise directed by the Employer.

The boring log is the basic record of geotechnical exploration and provides a detailed record of the work performed and the findings of the investigation. The Contractor shall prepare the field log in a legibly and clean format. All appropriate portions of the logs should be completed in the field prior to completion of the field exploration. The boring log, core boring log, test pit log forms and legends should be approved by the Employer before the commencement of the site investigation.

7.2 Field Supervision

The Contractor shall provide sufficient engineers or geologists full time in the field during exploration activities. The specialist shall supervise and observe all subsurface exploration operations including sample labeling and storage/preservation, classify samples and prepare logs of borings, soundings and pits.

7.3 Soil Exploration

7.3.1 Soil Drilling

The Contractor shall nominate the appropriate tools and equipment for soil drilling subject to the Employer's approval. The Contractor shall obtain quality soil samples. Particular care shall be exercised to properly remove all slough or loose soil from the boring before sampling. The Contractor shall ensure the stability of the sidewalls and bottom of the boring in soft clays or cohesionless soils below groundwater level.

7.3.1.1 Solid Stem Continuous Flight Augers

Solid stem continuous flight auger drilling is generally limited to stiff cohesive soils where the boring walls are stable for the entire depth of the boring. A drill bit is attached to the leading section of flight to cut the soil. The flights act as a screw conveyor, bringing cuttings to the top of the hole. As the auger drills into the earth, additional auger sections are added until the required depth is reached. The Contractor shall regularly monitor the cuttings and the reaction of the drilling equipment to identify stratification changes between sample locations.

7.3.1.2 Hollow Stem Continuous Flight Augers

In general hollow stem augers are very similar to the continuous flight auger except it has a large hollow center. When the hole is being advanced, a center stem and plug are inserted into the hollow center of the auger. The center plug with a drag bit attached and located in the face of the cutter head aids in the advancement of the hole and also prevents soil cuttings from entering the hollow-stem auger. The Contractor shall adopt Hollow-stem auguring methods in clay soils or in granular soils above the groundwater level.

7.3.1.3 Percussion Drilling

This method can be adopted in soil with gravel and boulders when the boring has to be advanced at a faster rate. This method consists of breaking of the strata by repeated blows from a chisel or drilling bit and bailing out the debris at intervals by adding water into the borehole. This method is not suitable for careful and very reliable sampling

operation because of the disturbance caused during boring. This method shall not be adopted unless otherwise specified or permitted by the Employer.

7.3.1.4 *Rotary Mud Circulation Drilling*

This method can be used in all types of soil below water level. In this method boring shall be conducted by rotating the bit fixed at the bottom of the drill rod. Proper care shall be taken to keep a firm contact between the bit and the bottom of the borehole. Bentonite or mud laden fluids shall be used as the drilling fluid to serve as the protective surface inside the borehole.

7.3.1.5 *Hand Auguring*

The Contractor shall carry out the hand auguring boring to obtain the shallow sub-surface information from Sites with difficult access or terrain where vehicle accessibility is not possible. The Contractor shall choose the appropriate type of hand augers based on the soil layers encountered during the investigation. The Contractor shall not advance the hand augers more than 8 m below the ground level. The Contractor shall prepare a field logs based on the visual examination of hand auger samples. The field logs shall include the nature, color, consistency and visual Classification of the soil, thickness of soil strata.

7.3.2 ***Test Pits***

The location and depth of test pits to be executed shall be specified in the Geotechnical Investigation Scope of Work or as amended and approved by the Employer.

7.3.2.1 *Shallow Test Pits*

The Contractor shall mobilize the appropriate non-heavy equipment (e.g. hand tools, backhoe) for the excavation of shallow test pits, i.e. excavated up to a maximum depth of 1 m to 2 m below the natural ground level. The Contractor shall modify the exploration depth in areas with expected high groundwater levels or weak soil layers and inform the Employer as soon as practicable.

7.3.2.2 *Dozer Trenches*

The Contractor shall mobilize the appropriate mechanical equipment for trench excavation. Trench shall be excavated up to a maximum depth of 5 m below ground level. The Contractor shall modify the exploration depth in areas with expected high groundwater levels or weak soil layers and inform the Employer as soon as practicable.

The Contractor shall maintain the level of the bottom of the pit so that each lift represents a uniform horizon of the deposit during excavation. At the surface, the Contractor shall place the excavated material in a safe and orderly manner adjoining the pit with separate stacks to identify the depth of the material. The Contractor shall backfill the exploration pits with the spoils generated during the excavation. The Contractor shall compact the backfilled material to avoid excessive settlements.

The Contractor shall prepare a log on the visual examination of soil strata. The log shall include the nature, color, consistency and visual Classification of the soil, thickness of soil strata, thickness of expansive soil and ground water level (if any), etc.

7.3.3 Soil Samples and Samplers

7.3.3.1 General

Sufficient number of soil samples shall be collected for reliable estimation of soil properties. The samples collected shall be either disturbed or undisturbed. Disturbed soil samples shall be collected for field identification and conducting laboratory tests such as sieve analysis, index properties, specific gravity, chemical analysis, etc. Undisturbed samples shall be collected to estimate physical, strength and settlement/deformation properties of the soil.

All tools, equipment and material required for sampling and the method of sampling shall conform to pertinent standards referenced in Section 3. All disturbed and undisturbed samples collected in the field shall be classified at Site as per relevant standards referenced in Section 3.

All samples shall be identified with the project number, date, borehole or test pit number, depth of sampling, etc. An arrow mark pointing towards the top surface of the undisturbed sample as the soil was In-Situ shall be made. Care shall be taken to prevent damage to the undisturbed soil samples during collection and transportation. The tube samples shall be properly trimmed at both ends immediately after extracting the samples from the borehole/test pit and suitably capped on both sides.

When the Contractor fails to collect undisturbed soil sample at a specified depth, the borehole shall be advanced by 0.50 m and shall be performed with a standard penetration test. The reason for not obtaining the undisturbed soil sample shall be indicated in the bore-log.

Precaution shall be taken to ensure that there shall not be any change in moisture content and disturbance of the soil samples and they shall be placed in a temporary store at the end of the day's work. All the samples shall be kept over a bed of sand, jute bags, saw dust, etc. and covered over the top with similar material. The bed and top cover shall be kept moist till they are properly packed in wooden boxes. The Contractor shall be responsible for packing and transporting of all the samples from Site to the laboratory within seven days after sampling with proper protection against loss and damage.

All samples shall be suitably packed in wooden boxes using sand, saw dust, etc. all around the samples before transporting to the laboratory for testing.

7.3.3.2 Disturbed Samples

Disturbed soil samples shall be collected from cuttings and from split spoon sampler in boreholes at regular intervals to provide complete description of soil profile and its variation. The samples shall be immediately stored in airtight jars or polythene bags and labeled with borehole/test pit number and depth.

The Contractor shall collect sufficient bulk samples in elevated areas in order to classify the material properties for the purpose of identifying the quality and quantity of the

material. Disturbed samples weighing about 250N shall be collected at shallow depths and immediately stored in polythene bags. The bags shall be sealed properly and shall be kept in wooden boxes or plastic containers.

7.3.3.3 Undisturbed Samples

In each borehole undisturbed sample (UDS) shall be collected, whenever possible, at regular intervals of 3 m and/or as directed by the Employer. The starting depth shall be staggered in alternate boreholes. Samples shall be collected in such a manner that the structure of soil and its moisture content do not get altered. The specification for the accessories required for sampling and the sampling procedure , as well as the procedure for obtaining undisturbed samples in cohesive and/or cohesionless soils shall comply with ASTM D7015.

7.4 Rock Coring

Rock cores are samples of record of the existing subsurface conditions at given borehole locations. The samples are expected to yield significant indications about the geological, physical, and engineering nature of the subsurface for use in the design and construction of an engineered structure.

Coring in rock shall be completed at specified locations or as per the Employer's instruction, in accordance with ASTM D2113.

7.5 Borehole and Test Pit Logs

The borehole and test pit log is the basic record of the geotechnical exploration and provides a detailed record of the work performed and the findings of the investigation. The Contractor shall prepare the field log in a legibly and clean format. All appropriate portions of the logs shall be completed in the field prior to completion of the field exploration. The boring log, core boring log, test pit log forms and legends shall be submitted for approval by the Employer before the commencement of the site investigation.

The following information shall be included in the boring logs:

- Project information
- Date of site investigation
- Topographic survey data including boring location and surface elevation, and bench mark location and datum, if available
- An accurate record of any deviation in the planned boring locations
- Identification of the sub-soils and bedrock including density, consistency, color, moisture, structure, geologic origin
- The depths of the various generalized soil and rock strata encountered
- Sampler type, depth, penetration, and recovery

- Sampling resistance in terms of hydraulic pressure or blows per depth of sampler penetration. Size and type of hammer. Height of drop.
- Soil sampling interval and recovery
- Rock core run numbers, depths & lengths, core recovery, and Rock Quality Designation (RQD)
- Type of drilling operation and equipment used to advance and stabilize the hole
- Comparative resistance to drilling
- Loss of drilling fluid
- Water level observations with remarks on possible variations due to tides and river levels.

8. IN-SITU TESTING

The Contractor shall perform the appropriate In-Situ tests to define the geostratigraphy, get samples and obtain direct measurements of soil properties and geotechnical parameters as per the project specifications and directions of the Employer.

8.1 Standard Penetration Test (SPT)

This test shall be conducted in all types of soil deposits met within the borehole to find the variation in the soil stratification by correlating with the number of blows required for unit penetration of a standard penetrometer, in accordance with ASTM D1586.

8.2 Cone Penetration Test (CPT)

The Cone Penetration Test (CPT) is an In-Situ test that provides continuous profiling of geostratigraphy and soil properties evaluation. The test shall be performed in accordance with ASTM D3441.

8.3 Vane Shear Test (VST)

Field Vane Shear Test (as applicable) shall be performed inside the borehole to determine the shear strength of cohesive soils, especially of soft and sensitive clays which are highly susceptible to sampling disturbance. The test shall be performed in accordance with ASTM D2573.

8.4 Dynamic Cone Penetrometer Test (DCP)

Dynamic Cone Penetrometer Test (as applicable) shall be used to assess the compaction effort of compacted materials, in accordance with ASTM D7380 or ASTM D6951.

8.5 Flat Plate Dilatometer Test (DMT)

Flat Plate Dilatometer Test (as applicable) shall be used for evaluation of Site stratigraphy, homogeneity, depth to firm layers, voids or cavities, and other discontinuities for use in the design of earthworks and foundations. The test shall be performed in accordance with ASTM D6635.

8.6 Pressuremeter Test (PMT)

This test (as applicable) shall be conducted in boreholes at desired depth to obtain the In-Situ stress-strain characteristics and strength parameters of soils layer by measuring the deformation of the probe at different pressures of the volumemeter. The test shall be performed in accordance with ASTM D4719.

8.7 Percolation Test (PT)

The pressure pulse test (percolation) method covers a field procedure for determining the transmissivity and storativity of geological formations with low-permeability. The test provides a good approximation of the capacity of the zone of interest to transmit water, if the test intervals are representative of the entire zone and the surrounding rock is fully water saturated. The test shall be performed in accordance with ASTM D4631.

Permeability can also be determined using Pump in tests and pump out tests and Packer testing in accordance with drilling industry standards.

8.8 Geophysical Methods

8.8.1 Electrical Resistivity Test (ERT)

This test (as applicable) shall be conducted to determine the electrical resistivity of soil required for designing safety grounding system for the entire power plant area. The test shall be performed in accordance with ASTM G57.

8.8.2 Seismic Refraction Test (SRT)

This test (as applicable) shall be carried out to establish the rock and soil profiles of varying density, as well as the dynamic shear modulus of the soil. The test shall be performed in accordance with ASTM D5777.

8.8.3 Cross Hole Seismic Test (CHT)

The seismic crosshole method (as applicable) shall be carried out to gather information pertinent to the seismic wave velocities of the materials in question. This data shall be used as input into static/dynamic analyses, as a means for computing shear modulus, Young' modulus, and Poisson' ratio, or for the determination of anomalies that might exist between boreholes. The test shall be performed in accordance with ASTM D4428.

9. GROUNDWATER INVESTIGATIONS

Groundwater conditions and the potential for groundwater seepage are fundamental factors in all geotechnical analyses and design studies. The Contractor shall record any groundwater conditions encountered during the geotechnical investigations. Groundwater investigations are of two types as follows:

- Determination of groundwater levels and pressures
- Measurement of the permeability of the subsurface materials.

9.1 Determination of Groundwater Levels and Pressures

The Contractor shall determine the groundwater levels as per ASTM D5092. Alternative

methods for groundwater monitoring can be adopted if approved by the Employer.

Either stand pipe or piezometer shall be installed at the specified depths in the selected previously drilled boreholes or specially drilled boreholes for this purpose as per the specification and instructions of the Employer.

9.2 Field Measurement of Permeability

In-situ permeability test shall be conducted to determine the water percolation capacity of overburden soil. This test shall be performed inside the borehole/test pit at specified depths or in each layer or as per the directions of the Employer.

10. LABORATORY TESTING

10.1 Essential Requirements

All laboratory tests shall be conducted in an approved laboratory using approved apparatus complying with relevant standards as referenced in Section 3.2. The Contractor shall ensure that the laboratory equipment is in good working condition before starting the laboratory tests. Calibration of all the instruments and their accessories shall be done carefully and precisely and as per the manufacturer recommendations.

Depending on the type of sub strata encountered, appropriate laboratory tests shall be conducted on soil and rock samples collected in the field. Laboratory tests shall be scheduled and performed by qualified and experienced personnel who are thoroughly conversant with the work. Tests indicated in the schedule of items shall be performed on soil, rock and water samples as per relevant ASTM standards. Copies of the laboratory test certificates shall be submitted to the Employer immediately by electronic means with follow-up hard copies. Laboratory tests shall be carried out concurrently with field investigation since initial laboratory test results could be useful in planning the later part of field work. The Contractor shall submit a Schedule of Laboratory Test to the Employer for approval.

The Contractor shall determine the requirement for chemical tests and other specific testing and shall include the proposed tests as part of the Schedule of Laboratory Test.

10.2 Laboratory Tests

Laboratory tests as indicated in this specification and approved by the Employer shall be conducted. Typical soil laboratory tests for soil samples are summarized below. The Contractor shall determine the requirement for specialized testing and shall include the proposed tests as part of the Schedule of Laboratory Test.

- Visual and engineering Classification
- Gradation analysis (sieve analysis and hydrometer analysis)
- Liquid, plastic and shrinkage limits
- Specific gravity

- Chemical analysis
- Swell pressure and free swell index determination
- Proctor compaction (standard and/or modified, as required)
- California bearing ratio
- Bulk density and moisture content
- Relative density (for sand)
- Unconfined compression
- Direct shear
- Permeability
- Shear (depending on the type of soil and field conditions on undisturbed or remoulded samples), and
- One dimensional consolidation.

Typical laboratory tests for rock samples are summarized as follows:

- Visual Classification
- Water absorption, porosity and density
- Specific gravity
- Hardness
- Slake durability
- Unconfined compression (both at saturated and at In-Situ water content)
- Point load strength index
- Deformability (both on saturated and dry samples)
- Direct tension
- Brazilian
- Ring shear
- Flexural, and
- Shear.

11. REPORT

11.1 General

On completion of all the field and laboratory works, the Contractor shall submit a draft report containing geological information of the region, procedure adopted for investigation, field observations, summarized test data, conclusion and recommendations. The executions of the investigation, conclusions and recommendations shall be based on the project requirements. The report shall include detailed boreholes logs, sub-soil sections, field test results, laboratory observations and test results in both tabular as well as graphical forms, practical and theoretical considerations for the interpretation of test results, the supporting calculations for the conclusions drawn. The Contractor shall submit as per the Scope of Work document the hard copies and electronic copy of the report in draft form for the Employer's review.

After review of the draft report, the Employer's comments shall be intimated to the Contractor. The Contractor shall incorporate the comments. After the amended draft report is approved, hard copies and electronic copy of the detailed final report shall be submitted as per the Scope of Work document.

11.2 Content of Report

The report shall include, but not be limited to the following:

- A plot plan showing the location and reduced levels of all field tests, e.g. boreholes, test pits, plate load tests, etc., properly drawn to scale and dimensioned with reference to the established grid lines. It shall show the project layout including main structures, roads, major infrastructure, both existing and proposed, and expected loading for each area/structure (if required).
- Geological information of the area such as geomorphology, geological structure, lithology, stratigraphy and tectonic faults, seismicity of the region and Site, core recovery and rock quality designation, etc
- Available past observations and historical data for the area or for other areas with similar soil profile for similar structures in the surrounding areas
- A cross section of all individual boreholes and test pits with levels and coordinates showing the Classification and thickness of individual stratum, position of ground water levels, various In-Situ tests conducted and samples collected at different depths, and the rock stratum (if encountered)
- A set of longitudinal and transverse soil/rock profiles connecting various boreholes shall be included to show variation of the subsoil strata as per relevant standards referenced in Section 3.2. The Contractor shall determine the number of profiles required in order to clearly show the subsoil strata variations subject to the Employers approval.
- Water level contours and rock level contours

- Plot of standard penetration test 'N' values (both uncorrected and corrected) and CPT values with depth for the identified areas
- A summary of typical soil profile indentified at the Site
- Results of all field tests in tabular as well as in graphical forms. Results of all laboratory tests summarised for each sample and a consolidated table giving the layer-wise soil and rock properties. All the relevant charts, tables, graphs, figures, supporting calculations, conclusions and photographs of representative rock cores and clay samples and test pits shall be furnished.
- For all shear tests, stress versus strain/deformation diagrams as well as Mohr's circle envelopes shall be furnished. If back pressure is applied for saturation, the magnitude of the same shall be indicated. The value of modulus of elasticity 'E' shall be furnished for all tests along with relevant calculations.
- For all consolidation tests the following curves shall be furnished: e versus log P, e versus P and compression versus log t or square root of t (depending upon shape of the plot for proper determination of coefficient of consolidation). The point showing initial condition (e_o, P_o) of the soil shall be marked on the curves. Values of compression index, coefficient of volume compressibility, etc. shall be also furnished. The procedure adopted for calculating the compression index from the field curve and settlement of soil strata shall be clearly specified. The time required for 50% and 90% primary consolidation along with secondary settlement (if significant) shall also be calculated.
- For Pressuremeter tests, the data shall be presented as per the applicable codes and standards
- Values of cohesion, angle of internal friction, pressuremeter modulus, shear modulus and co-efficient of sub-grade reaction along with sample calculations. Calculation for allowable bearing pressures and corresponding total settlements for shallow foundations and load carrying capacity calculation of piles in various modes, etc., as per relevant standards referenced in Section 3.
- Analysis and discussion of test results
- If the design of ground engineering structures such as retaining walls, basements, tunnels, and/or deep foundations, etc. is required, lateral earth pressure coefficients 'K' (at rest, active and/or passive) shall be provided and used to describe the soil pressure.

11.3 Recommendations

Recommendations shall be given area wise duly considering project requirements, the type of soil/rock, structure, foundation type and ground water table, etc. in the area. The recommendations shall include, as applicable, but not be limited to the following:

- Type of foundation to be adopted for various structures duly considering the sub-strata characteristics, water table, total settlement permissible for the structures and equipments, minimum depth and width of foundation, etc
- For shallow foundations the following shall be indicated with comprehensive supporting calculations:
  - Net safe bearing pressure for isolated circular (if expected), square or rectangular footings and continuous strip footings of sizes 1, 2, 3, 4 & 5 m at different founding depths of 1, 2, 3, 4 & 5 m below ground level and depths expected for certain structures, considering both shear failure and settlement criteria giving reasons for the type of shear failure adopted in the calculation
  - Net safe bearing pressure for raft foundation of widths greater than 6 m at 2, 3, 4 & 5 m below ground level and depths expected for certain structures, considering both shear failure and settlement criteria
  - Modulus of sub-grade reaction and modulus of elasticity from plate load test results and from correlations with SPT and CPT tests results and other similar tests, along with time-settlement curves and load settlement curves in both natural and log-log graph.
- If piling is envisaged the following shall be furnished with comprehensive supporting calculations:
  - Type of pile and reasons for recommending the same duly considering the sub-strata characteristics and loading conditions
  - Suitable founding strata for the type of pile considered
  - Estimated length of pile and pile capacities. End bearing and frictional resistance shall be indicated separately. Safe lateral and tensile load carrying capacities of pile with supporting calculations.
  - Calculation of the magnitude of negative skin friction (if any), in case of friction and end bearing piles where the pile receives its principal vertical support from skin friction between the surface of the buried pile and the surrounding soil.
- Coefficient of permeability of various sub-soil and rock strata based on In-Situ permeability tests (if conducted), or correlations with other tests
- Cone resistance, frictional resistance, total resistance and settlement analysis for different size of foundations
- Standard penetration test results, if conducted or correlated with other tests, to estimate the resistance of soil to penetration
- Lateral earth pressures and coefficients used in the analysis depending on the

characteristics of the design (if applicable)

- Electrical resistivity of sub-soil based on electrical resistivity tests including electrode spacing versus cumulative resistivity curves (if any)
- Dynamic soil properties such as dynamic shear modulus and Poisson's ratio, etc. from cross-hole seismic refraction tests and coefficient of elastic uniform compression from cyclic plate load tests (if any)
- Suitability of the soil for construction of roads and embankments, their stable slopes for shallow and deep excavations and embankments, active and passive earth pressures, earth pressure at rest and modulus of elasticity as a function of depths for the design of underground structures, etc
- Suitability of locally available soils on Site as borrow material. The Contractor shall supply details of the quality and quantity of the material available.
- Where problematic soils such as expansive soil or sabkha, etc. are encountered, recommendations on removal or detainment of the same under the structures/roads, etc. shall be given. In the latter case, detailed specifications of any special treatment required including specification for materials to be used, construction method, equipments to be deployed, etc. shall be furnished.
- Protective measures based on chemical nature of soil and ground water with due regard to the potential deleterious effects on concrete, steel and other building materials. Remedial measures for sulphate attack and acidity shall be dealt with in detail.
- Susceptibility of sub soil strata to liquefaction in the event of earthquake. If so, recommendation for remedial measures shall be included.
- Identification of any other potential geotechnical problems & their remedial measures
- Description of measures required for erosion control
- Identification of corrective measures required for the improvement of sub surface conditions such as removal of poor sub soil/material and In-Situ densification, etc. If ground improvement is recommended, its detailed specification, specification for the materials to be used, construction method, equipments to be deployed and quality control, etc. shall be furnished.

12. QUALITY CONTROL

The Contractor shall refer to the following:

MD-101-SMPM-PM-QA-PEI-0001 Contractor Quality Requirements